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1) How java works internally.

→ ① The user write a code in a .java file and when the user tries to run the .java file it gets compiled by the java compiler with the help of javac application and checks the syntax error by using .jar files.

② if there is no errors then the source code (.java) file will convert it into .class ~~file~~ file (.bytecode).

③ Then the byte code is sent to the JVM. where it has its interpreter as per their specific OS.

<u>Note</u>:- Each ~~os has~~ JVM has its own ~~os~~ interpreter for their specific OS.

④ The interpreter will read the code line by line & translate code from byte code to machine code.

2) What is JDK & JRE.

→ JDK is a java development kit it is a collection of all the required needs to develope a project for a developer e.g:- compiler, debugger, documentation tools & JRE.

JRE Java Runtime Environment. It is used to run or execute the

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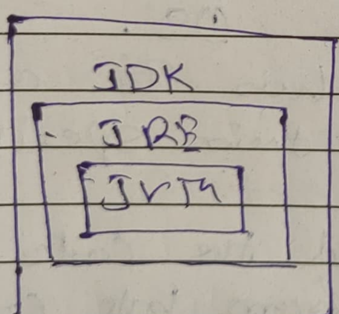
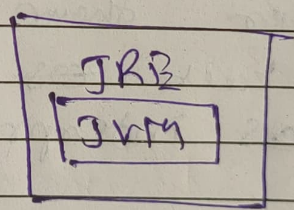
java code or an application. It has the JVM where the interpreter is located. the JRE will test run the code ~~is~~ before the end user. the JVM contains the other standard libraries that are required to run a code or program.

In simple terms the application downloaded from ~~[illegible]~~ a playstore is a compiled program in java/byte code and as per the user we need only JRE in our device to run that application.



Developers used.



End users Needs.

3] what is the role of compiler & interpreter.

⇒ It is a collection of code / methods ~~de~~ which are present in a JDK. It is predefined.

The compiler has two functions.

1. Errors Detection.

The compiler uses .jar file as

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a reference to check for any system errors in a code.

2. Conversion.
It converts the Source Code into byte Code for process further in JRE.

<u>Interpreter</u> :- In simple words "translator" which translates ~~the~~ or converts the byte code (.class) into machine code (binary). The interpreter is located in JVM and JVM is located into JRE.

These are different interpreter as per the different types of OS.

4] what is JVM.
→ JVM (Java Virtual Machine) it has the interpreter which is used to convert ~~of~~ byte code into machine code. The JVM is responsible for executing the code on all existing OS.

Note:- JVM is ~~perm~~ partically platform independent as different os required different interpreter.

5] why java is called as platform independent.

⇒ • The java is called as ~~platf~~ platform independent because the programmer write a code in java which is conversed into byte code and the byte code can run on any operating system which has JVM installed.

Each JVM is made as per their specific OS. But the byte code can run on any OS as ~~as~~ long as it has JVM installed.

6] why to create main ~~the~~ method in test class.

⇒ The main () method is the start point of the execution. without the main method the program cannot run. JVM initially looks for the main () method to know from where to start.

The developers will write the logic of a code is seperatelly to the main() method where the main ~~method~~ method will call all the ~~method~~ methods / functions.